AYMANE AIT DADS

AI Compiler & Inference Software Engineer Intern | Neural Network Optimization · MLIR Runtime · PyTorch Integration

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, $106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by designing and optimizing a full LoRA SFT training pipeline on a 30B-parameter Mamba-Transformer, demonstrating deep command of inference-critical model behavior and hardware-aware training design. Fine-tuned a CLIP ViT-L/14 (428M parameters) inference pipeline achieving 1st-place AUC in class and a submission to NTIRE 2026 at CVPR, applying FP16 mixed precision, ensemble inference, and test-time augmentation to maximize deployment performance. Brings hands-on experience with PyTorch, BF16/FP16 mixed-precision execution, GPU debugging, Triton, and vLLM - directly applicable to co-designing compiler and runtime features that extract maximum performance from custom AI hardware. Aims to contribute to Tesla's MLIR-based AI inference stack for FSD and Optimus by applying rigorous ablation methodology and low-level optimization insight to compiler and runtime development challenges.

CORE COMPETENCIES

AI Inference Stack | Compiler / Runtime Development | Neural Network Optimization | PyTorch Framework Integration | Mixed-Precision Execution (BF16/FP16) | MLIR & Code Generation | Distributed Inference Design | Ablation & Performance Analysis

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered an automated **feature engineering and model evaluation pipeline** in Python and Scikit-learn on 100K+ fiber-optic network records, reducing manual analysis time by ~60%.
- Built a Random Forest classifier and K-Means clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, outputs adopted directly by the network optimization team.
- Developed clustering models processing upload speed, latency, and jitter signals across **thousands of network nodes**, enabling quantitative quality classification at scale.
- Delivered stakeholder presentations translating model outputs into maintenance recommendations, communicating **technical findings** across engineering and business teams using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)
Kaggle Competition · June 2026 · Ranked 17th / 3,000+ Teams · $106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) targeting all attention projections, MLP layers, and lm_head across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that over-expanded reasoning chains - not incorrect reasoning - was the primary failure mode in hard categories.

PyTorch · Unsloth · TRL · PEFT · vLLM · Triton · Hugging Face Transformers · Kaggle GPU

AI-Generated Image Detection - NTIRE 2026 @ CVPR
EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in EURECOM Class (0.791 AUC)

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head, dual learning rates, and FP16 mixed-precision training on 250K images from 25+ generator types; also submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Discovered that **1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC)** for out-of-distribution generalization; deployed a 3-model ensemble weighted by validation AUC with 10-view test-time augmentation to maximize inference robustness.

PyTorch · OpenCLIP · ViT · AdamW · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

AI Compiler & Inference Engineering: PyTorch | BF16/FP16 Mixed Precision | vLLM | Triton | CUDA Debugging | Unsloth | TRL

LLM Fine-Tuning & Model Optimization: LoRA | PEFT | SFTTrainer | Completion-Only Loss | Mamba-Transformer | Hugging Face Transformers | Ablation Studies

Systems, MLOps & Frameworks: Python | Bash | Git | Linux | Docker | PostgreSQL | Scikit-learn | Pandas